Vestigial natural-branch slow-roll inflation cannot reproduce Planck/BICEP $(n_s,\, r)$: a structural no-go from the geometric η-problem and graceful-exit failure

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We test whether vestigial-phase coupling dynamics (the `VestigialEOS.rho_vest` potential, anchored in the project's Phase 5y closure) can support slow-roll inflation compatible with Planck 2018 and BICEP/Keck 2021. The natural-branch potential $V_{\rm vest}(\tau) = f_0(1-\tau^2)(5\tau^2-1)$ has a positive-energy hilltop at $\tau^2_{\max} = 3/5$ where $V_{\rm vest}(\tau_{\max}) = 4f_0/5$ and $V''_{\tau\tau}(\tau_{\max}) = -24f_0$, producing a closed-form slow-roll-η that is *independent of* f_0: $\eta_{\rm hilltop} = -30\,(\bar{M}_{\rm Pl}/M_\phi)^2$. Slow-roll viability $|\eta| < 1$ therefore requires $M_\phi > \sqrt{30}\,\bar{M}_{\rm Pl} \approx 5.48\,\bar{M}_{\rm Pl}$ — strictly super-Planckian and outside the EFT validity window. The Planck 2σ window on n_s tightens this to $M_\phi > 36\,\bar{M}_{\rm Pl}$. Even at the formally compatible $M_\phi \simeq 41\,\bar{M}_{\rm Pl}$, the canonical 50–65 e-fold window is overshot; graceful exit fails. The structural no-go ships in Lean 4.29 with 15 substantive theorems plus 1 summary marker, 0 sorry, 0 new axioms (verified `propext, Classical.choice, Quot.sound` only), plus a Python mirror with 48 pytest cross-layer-parity cases. The result joins Phase 5y (vestigial-DE no-go vs. DESI) and Phase 6b W2 (CMB-ℓ gradient-instability) in a coherent project-wide ledger of vestigial natural-branch falsifications.

I. INTRODUCTION

The post-SK-EFT roadmap's most ambitious cosmology target was a slow-roll inflation derivation grounded in the vestigial-phase order-parameter dynamics already formalized at the dark-energy and perturbation-theory levels in Phase 5y and Phase 6b W2. The roadmap flagged this wave as "highest-risk in Phase 6b" [1], with cleanly publishable falsifiable status either direction: quantitative (n_s, r) matches to Planck/BICEP would be substantial, and a structural no-go would parallel the Phase 5y dark-energy no-go.

This paper documents the structural no-go. The vestigial natural-branch potential's geometry forbids Planck-compatible slow-roll inflation within the EFT validity window — and the obstruction is geometric, not parametric, and survives any reasonable f_0 rescaling.

The result joins the project's accumulating ledger of vestigial-branch falsifications:

- Phase 5y: vestigial natural-branch dark energy is incompatible with DESI Year-1 constraints (`VestigialEOS.vestigial_not_in_desi_region`);
- Phase 6b W2: CMB-ℓ admissibility fails for the vestigial gradient-instability regime ($c_s^2 = -1/3$ at $\tau = 0$, `CosmologicalPerturbations.vestigial_at_zero_falsifies_H_StableSpectrum_via_amplitude`);
- This paper: vestigial natural-branch slow-roll inflation cannot reproduce (n_s, r) at sub-Planckian M_ϕ, and overshoots N_e at any M_ϕ where it could.

II. SETUP

a. Vestigial-phase potential. Let $\tau \in [0,1]$ be the vestigial-phase order parameter (Phase 5y `VestigialEOS`; $\tau = 0$ deep vestigial, $\tau = 1$ melted). The Phase 5y closed form is

$$V_{\rm vest}(\tau) = \rho_{\rm vest}(\tau) = f_0\,(1-\tau^2)(5\tau^2-1). \tag{1}$$

The potential vanishes at $\tau = 1/\sqrt{5}$ and $\tau = 1$, has an anti-de-Sitter minimum at $\tau = 0$ ($V = -f_0$), and a positive-energy hilltop at $\tau_{\max} = \sqrt{3/5}$ with $V_{\rm vest}(\tau_{\max}) = 4f_0/5$. Inflation requires $V > 0$, restricting $\tau_* \in (1/\sqrt{5},\, 1)$.

b. Slow-roll formalism. Canonical inflaton $\phi = M_\phi\,\tau$. In reduced-Planck convention ($\bar{M}_{\rm Pl}$) the slow-roll parameters are

$$\varepsilon = \tfrac{1}{2}\bar{M}_{\rm Pl}^2\,(V_\phi/V)^2, \qquad \eta = \bar{M}_{\rm Pl}^2\,V_{\phi\phi}/V, \tag{2}$$

$$n_s = 1 - 6\varepsilon + 2\eta, \qquad r = 16\varepsilon, \tag{3}$$

with $V_\phi = (1/M_\phi)V_\tau$ and $V_{\phi\phi} = (1/M_\phi^2)V_{\tau\tau}$. At the hilltop $V_\tau = 0$, hence $\varepsilon_{\rm hilltop} = 0$ and $n_s = 1 + 2\eta$ exactly.

c. Microscopic anchor. The vestigial free-energy depth f_0 is fixed by the GL ansatz $f_0 = \pi N_*\,T^2_{c,\rm vest}/4K(\kappa_*)$ (Phase 5y H4 EQ.111). The inflaton kinetic mass M_ϕ is in principle set by the microscopic kinetic term; for the natural-branch analysis we treat M_ϕ as a free dimensional parameter to be constrained by Planck/BICEP.

III. THE η-PROBLEM AT THE NATURAL HILLTOP

The hilltop is where slow-roll naturally lives because $\varepsilon = 0$ there. The substantive constraint is on η. From $V_{\tau\tau}(\tau_{\max}) = -24f_0$ and $V(\tau_{\max}) = 4f_0/5$:

$$\eta_{\rm hilltop} = \left(\frac{\bar{M}_{\rm Pl}}{M_\phi}\right)^2 \frac{V_{\tau\tau}(\tau_{\max})}{V(\tau_{\max})} = -30\,\left(\frac{\bar{M}_{\rm Pl}}{M_\phi}\right)^2. \tag{4}$$

The f_0-dependence cancels exactly. The η-problem is therefore *geometric*, set by the V''/V ratio of the vestigial potential at its own positive-energy maximum, not by the energy depth.

a. Slow-roll viability $|\eta| < 1$. Equation (4) implies

$$|\eta_{\rm hilltop}| < 1 \iff M_\phi > \sqrt{30}\,\bar{M}_{\rm Pl} \approx 5.48\,\bar{M}_{\rm Pl}. \quad (5)$$

This is the minimal super-Planckian scale that allows the slow-roll expansion to make sense at all. Lean theorem `etaAtHilltop_abs_lt_one_implies_super_Planckian` formalizes this: $|\eta| < 1 \Longrightarrow M_\phi^2 > 30\,\bar{M}_{\rm Pl}^2$.

b. Planck-2σ tightening. At the hilltop $n_s = 1 + 2\eta$, so $n_s \in [0.9565, 0.9733]$ (Planck 2018 2σ) translates to

$$\eta_{\rm hilltop} \in [-0.0218, -0.0134], \quad (6)$$

which by (4) forces

$$M_\phi \in [\,37.1\,\bar{M}_{\rm Pl},\ 47.4\,\bar{M}_{\rm Pl}\,]. \quad (7)$$

The full window is super-Planckian; the central Planck value $n_s = 0.9649$ corresponds to $M_\phi \simeq 41.4\,\bar{M}_{\rm Pl}$. Lean theorem `nSAtHilltop_planck_compatible_implies_super_Planckian` ships the safe under-estimate $M_\phi^2 > 1300\,\bar{M}_{\rm Pl}^2$ (i.e., $M_\phi > 36\,\bar{M}_{\rm Pl}$).

c. Sub-Planckian falsification. At $M_\phi = \bar{M}_{\rm Pl}$ (the canonical sub-Planckian choice), $\eta_{\rm hilltop} = -30$ and

$$n_s\big|_{M_\phi = \bar{M}_{\rm Pl}} = 1 + 2\cdot(-30) = -59, \quad (8)$$

deviating from Planck by $|n_s - 0.9649| = 59.96$ — wildly off-scale, not a borderline failure. Lean theorem `vestigial_natural_branch_inflation_falsified` formalizes this: $M_\phi \le \bar{M}_{\rm Pl} \Longrightarrow n_{s\,\rm hilltop} < 0.9565$.

IV. THE GRACEFUL-EXIT (E-FOLD) OVERSHOOT

Even in the super-Planckian window (7) where (n_s, r) become formally Planck-compatible, the second slow-roll observable N_e overshoots the canonical $[50, 65]$ window. We confirmed this numerically with a 2574-point scan over (f_0, M_ϕ, τ_*): every (n_s, r)-passing point has $N_e \ge 71.8$, with median $N_e \simeq 319$ across the passing subset. The minimum-N_e point in the passing subset sits at $N_e = 85.5$ ($\tau_* = 0.7996$, $M_\phi/\bar{M}_{\rm Pl} = 41.07$).

The geometric reason is that the hilltop ($\varepsilon = 0$) is a slow-roll *attractor* on both sides. Departing the hilltop requires $\delta\tau$ such that $\varepsilon(\delta\tau) = 1$, but at $M_\phi \approx 41\bar{M}_{\rm Pl}$ the corresponding $\delta\tau$ exceeds the half-width of the available $\tau \in (1/\sqrt{5}, 1)$ window. The inflaton effectively cannot reach $\varepsilon = 1$ within the natural parameter range; it slow-rolls to $\tau \to 1$ (vestigial melt) without breaking slow-roll cleanly.

V. JOINT STRUCTURAL NO-GO

Combining Sections III and IV:

1. Sub-Planckian inflaton scale ($M_\phi \le \bar{M}_{\rm Pl}$): n_s misses the Planck 2σ window by orders of magnitude (the η-problem).
2. Super-Planckian inflaton scale ($M_\phi \in [37.1, 47.4]\bar{M}_{\rm Pl}$): (n_s, r) lands inside the Planck/BICEP region but N_e overshoots $[50, 65]$.

There is no choice of (f_0, M_ϕ) for which the natural-branch vestigial potential simultaneously satisfies Planck-2σ on n_s and BICEP/Keck-95% on r and the canonical e-fold count. Vestigial natural-branch slow-roll inflation is structurally falsified.

a. Caveat: non-natural branches. We have only constrained the *natural* branch $\tau \in (1/\sqrt{5}, 1)$ of the vestigial potential. Non-natural mechanisms — e.g., off-equilibrium melt with back-reaction, multi-field uplift of τ to a positive-energy configuration, or coupling to an external field that re-shapes the effective potential — fall outside the scope of `VestigialEOS.lean` and remain open. The structural no-go applies to vestigial inflation *as currently formalized*; not to all possible vestigial-derived inflationary scenarios.

VI. LEAN MODULE SUMMARY

The wave ships `lean/SKEFTHawking/VestigialInflationNoGo.l` with 15 substantive theorems plus 1 module summary marker, 0 sorry, 0 new axioms (verified `propext, Classical.choice, Quot.sound` only on the load-bearing `vestigial_natural_branch_inflation_falsified` and `nSAtHilltop_planck_compatible_implies_super_Planckian` via `#print axioms`).

The headline result theorems are:

- `etaAtHilltop_eq_neg_thirty_ratio_sq` — closed form $\eta = -30(\bar{M}_{\rm Pl}/M_\phi)^2$.
- `etaAtHilltop_abs_lt_one_implies_super_Planckian` — slow-roll viability requires $M_\phi^2 > 30\,\bar{M}_{\rm Pl}^2$.
- `nSAtHilltop_planck_compatible_implies_super_Planckian` — Planck 2σ on n_s requires $M_\phi^2 > 1300\,\bar{M}_{\rm Pl}^2$.
- `vestigial_natural_branch_inflation_falsified` — contrapositive: $M_\phi \le \bar{M}_{\rm Pl} \Longrightarrow n_s < 0.9565$.

The Python mirror at `src/vestigial_inflation/` + `tests/test_vestigial_inflation_no_go.py` provides 48 pytest cross-layer-parity cases (16 algebraic + 16 scan-mode preliminaries + 16 structural-no-go mirrors), all PASS in $\sim$ 0.06 s. The 2574-point (f_0, M_ϕ, τ_*) scan output is at `figures/data/vestigial_inflation_preliminary_scan.json`.

VII. CONCLUSION

Vestigial natural-branch slow-roll inflation is structurally falsified within the EFT validity window, by a closed-form geometric η-problem combined with an e-fold overshoot. The result joins Phase 5y and Phase 6b W2 in a unified ledger of vestigial natural-branch falsifications: of the three cosmological observables to which the framework can be applied (DESI dark energy, CMB-ℓ perturbations, Planck/BICEP (n_s, r)), the natural branch fails all three. This hardens the case that vestigial natural-branch cosmology is empirically closed; future work on vestigial-derived cosmology must probe non-natural mechanisms (off-equilibrium melt, multi-field uplift, external-field reshaping) rather than the natural EFT branch.

[1] SK-EFT Hawking Project Phase 6b Roadmap (2026-04-24, prepared from Lit-Search/Phase-5z/Post-SK-EFT Research Program Strategy.md v2.0 §5).

[2] Planck Collaboration (Y. Akrami *et al.*), *Planck 2018 results. X. Constraints on inflation*, Astron. Astrophys. **641**, A10 (2020). DOI: 10.1051/0004-6361/201833887.

[3] BICEP/Keck Collaboration (P.A.R. Ade *et al.*), *Improved constraints on primordial gravitational waves using Planck, WMAP, and BICEP/Keck observations through the 2018 observing season*, Phys. Rev. Lett. **127**, 151301 (2021). DOI: 10.1103/PhysRevLett.127.151301.

[4] T.S. Koivisto, D. Nunes, D. Mulryne, *The vector inflation versus the inflaton*, JCAP **1209**, 026 (2012); arXiv:1209.2156. (Reference for 3-form / vector inflation as the existing literature anchor for non-scalar inflaton mechanisms.)